■ MDB Agent System

Installation & User Guide

A complete step-by-step guide for installing Python, setting up the Flask application, and using all system features.

Platform	Windows
Language	Html, CSS, Python
Framework	Flask (Web Application)
Database	SQLite (built-in)

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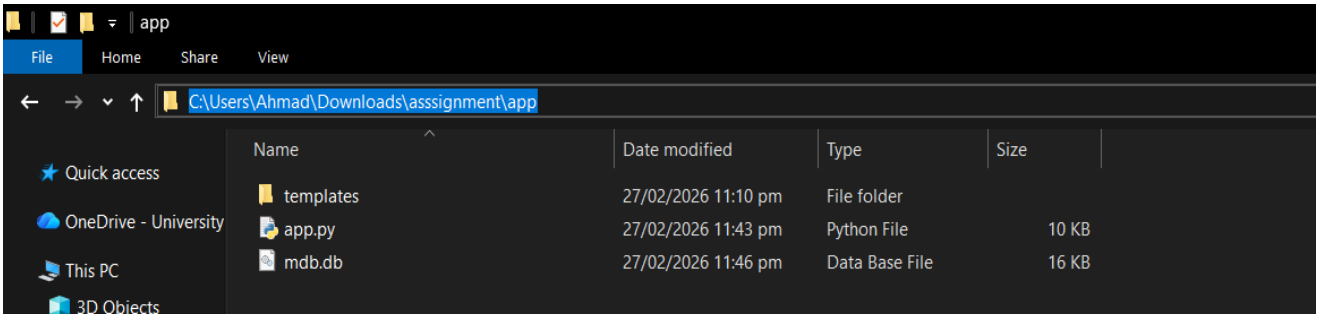
1. Installing Python

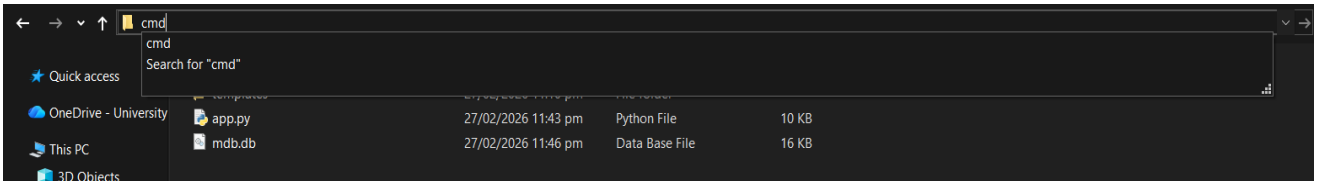
Windows

1. Open your browser and Download Python installer.
   https://www.python.org/ftp/python/3.14.3/python-3.14.3-amd64.exe
2. Run the downloaded installer (.exe file).
3. **Important:** On the first screen, check the box that says **"Add Python to PATH"** before clicking Install Now.
4. Click **Install Now** and wait for the installation to complete.
5. Click **Close** when done.

2. Installing Dependencies

Just run install flask file inside app directory it will auto install it.





3. Running the Application

After installing python and Flask just run start app file.

You should see output like:

```
* Running on
  http://127.0.0.1:5000 * Debug
  mode: on
* Restarting with stat
```

Open your web browser and go to: **http://127.0.0.1:5000** or **http://localhost:5000**

To stop the server, press **Ctrl + C** in the terminal or just close it.

First run note: On the first run, the application automatically creates the SQLite database file (**mdb.db**) and inserts a default administrator account. You do not need to set up the database manually.

4. First-Time Setup & Default Admin Account

A default administrator account is automatically created when the app runs for the first time:

Field	Value
Email	maryam@admin.com
Password	admin123
Role	Admin

5. Using the Application

5.1 Registration (Students)

New students can create their own accounts by navigating to the registration page.
Instructor and Admin accounts can only be created by an existing Admin.

To register as a student:

1. Go to **http://localhost:5000/register** or click **"Register as Student"** on the login page.
2. Fill in your **Full Name**, **Email Address**, **Password**, and **Confirm Password**.
3. Password must be **at least 6 characters** long.
4. Both password fields must match.
5. Click **Create Account**.
6. You will be redirected to the login page with a success message.

Note: Each email address can only be registered once. If you see an error saying "Email already registered", the address is already in use.

5.2 Logging In

All users (students, instructors, and admins) log in from the same login page:

1. Navigate to **http://localhost:5000** — you will be automatically redirected to login.
2. Enter your registered **Email Address** and **Password**.
3. Click **Sign In**.
4. You will be automatically redirected to the dashboard appropriate for your role.

If your account has been blocked by an administrator, you will see a message: "Your account has been blocked. Contact administrator." and will not be able to log in.

5.3 Student Dashboard

After logging in as a student, you are taken to the Student Dashboard at **/student/dashboard**. This page shows:

- **Statistics cards** showing enrolled courses, assignments, and pending tasks (currently placeholders).

- **My Profile** section showing your name, role badge, and account status.
- A **Logout** button in the top navigation bar.

Students have read-only access and cannot modify any data.

5.4 Instructor Dashboard

Instructors are redirected to **/instructor/dashboard** after login. This dashboard provides:

- **Statistics**: Total students, active students, and blocked students count.
- **Student List**: A read-only table showing all registered students with their name, email, status, and registration date.

Instructors have view-only access to student information. They cannot block, delete, or modify any accounts.

5.5 Admin Panel

Admins are redirected to **/admin/panel** after login. This is the most powerful section of the application. The admin panel is divided into three main sections:

Overview Stats

The top of the admin panel shows four statistics cards:

- **Total Users** (blue) – total number of accounts in the system.
- **Students** (green) – count of accounts with student role.
- **Instructors** (orange) – count of accounts with instructor role.
- **Blocked** (red) – count of blocked accounts.

Create New User

Admins can create accounts for any role. Fill in the form with:

- **Full Name** – the user's display name.
- **Email Address** – must be unique in the system.
- **Password** – the initial password for the new user.
- **Role** – select Student, Instructor, or Admin from the dropdown.

Click **Create User** to add the account. The new user will appear in the table below.

All Users Table

The users table lists every account in the system with the following columns:

Column	Description
#	The unique numeric ID of the user
Name	The user's full name
Email	The user's email address (login identifier)
Role	Dropdown to change role — save with the Set button
Status	Active (green) or Blocked (red)

Registered	Date the account was created
Actions	Block/Unblock button and Delete button

6. Admin Operations Reference

Block a User

Click the orange **Block** button next to a user. A confirmation prompt will appear. Blocked users cannot log in until unblocked. Admins cannot block themselves.

Unblock a User

When a user's status is Blocked, the Block button is replaced by a green **Unblock** button. Click it to restore the user's access immediately.

Delete a User

Click the red **Delete** button next to a user and confirm the prompt. This action is **permanent and cannot be undone**. Admins cannot delete their own account.

Change a User's Role

In the Role column of the users table, select a new role from the dropdown (Student / Instructor / Admin) and click **Set**. The change takes effect immediately. Admins cannot change their own role.

Create an Instructor Account

Use the **Create New User** form, enter the instructor's details, select **Instructor** from the Role dropdown, and click Create User.

Create Another Admin Account

Use the Create New User form and select **Admin** as the role. Use this carefully – admins have full system access.